MCA SEM-I

SUBJECT: Open-Source Web Programming Using PHP

UNIT- I

PRACTICAL LIST WITH SOLUTION

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Practical 1: PHP variables, data types, operators and control statements

Unit 1

Aim: To implement and demonstrate PHP variables, data types, operators and control statements.

Procedure:

Start XAMPP and Apache.

Create practical1.php inside C:\xampp\htdocs\.

Enter the program, save it, and open http://localhost/practical1.php in the browser.

Verify the output.

Solution:

```
<?php
$name="Rahul"; $age=22; $marks=78.5;
$total=356; $max=450;
$percentage=($total/$max)*100;
echo "Name: $name<br>";
echo "Age: $age<br>";
```

```
echo "Percentage: $percentage%<br>";
if($percentage>=75) echo "Grade: Distinction";
elseif($percentage>=60) echo "Grade: First Class";
elseif($percentage>=40) echo "Grade: Pass";
else echo "Result: Fail";
?>
```

Expected Output:

Name: Rahul | Age: 22 | Percentage: 79.11% | Grade: Distinction

Result:

Thus, the practical was executed successfully and the required result was obtained.

Viva Questions:

What is a PHP variable?

What is an operator in PHP?

What is an if-else statement?

Practical 2: PHP application to swap two numbers without using a third variable

Unit 1

Aim: To implement and demonstrate PHP application to swap two numbers without using a third variable.

Procedure:

Start XAMPP and Apache.

Create practical2.php inside C:\xampp\htdocs\.

Enter the program, save it, and open http://localhost/practical2.php in the browser.

Verify the output.

Solution:

```
<?php
$a=10; $b=20;
echo "Before: A=$a, B=$b<br>";
$a=$a+$b;
$b=$a-$b;
$a=$a-$b;
echo "After: A=$a, B=$b";
?>
```

Expected Output:

Before: A=10, B=20 | After: A=20, B=10

Result:

Thus, the practical was executed successfully and the required result was obtained.

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Viva Questions:

Why is a third variable not required?

Which arithmetic operations perform the swap?

Can this method be used with decimals?

Practical 3: User-defined functions and variable scope

Unit 2

Aim: To implement and demonstrate user-defined functions and variable scope.

Procedure:

Start XAMPP and Apache.

Create practical3.php inside C:\xampp\htdocs\.

Enter the program, save it, and open http://localhost/practical3.php in the browser.

Verify the output.

Solution:

```
<?php
$college="ABC College";
function square($n){ $result=$n*$n; return $result; }
function showCollege(){ global $college; echo "College: $college"; }
echo "Square of 6: ".square(6)."<br>";
showCollege();
?>
```

Expected Output:

Square of 6: 36 | College: ABC College

Result:

Thus, the practical was executed successfully and the required result was obtained.

Viva Questions:

What is a user-defined function?

What is local scope?

How is a global variable accessed inside a function?

Practical 4: String manipulation and regular expressions

Unit 2

Aim: To implement and demonstrate string manipulation and regular expressions.

Procedure:

Start XAMPP and Apache.

Create practical4.php inside C:\xampp\htdocs\.

Enter the program, save it, and open http://localhost/practical4.php in the browser.

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Verify the output.

Solution:

```
<?php
$text="MCA PHP Programming";
$email="student@example.com";
echo "Original: $text<br>";
echo "Uppercase: ".strtoupper($text)."<br>";
echo "Length: ".strlen($text)."<br>";
echo "Words: ".str_word_count($text)."<br>";
$pattern="/^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$/";
echo preg_match($pattern,$email)?"Valid Email":"Invalid Email";
?>
```

Expected Output:

Original: MCA PHP Programming | Uppercase: MCA PHP PROGRAMMING | Length: 19 | Words: 3 | Valid Email

Result:

Thus, the practical was executed successfully and the required result was obtained.

Viva Questions:

What does strlen() do?

What is a regular expression?

Which function checks a regex in PHP?

Practical 5: Indexed, associative and multidimensional arrays

Unit 3

Aim: To implement and demonstrate indexed, associative and multidimensional arrays.

Procedure:

Start XAMPP and Apache.

Create practical5.php inside C:\xampp\htdocs\.

Enter the program, save it, and open http://localhost/practical5.php in the browser.

Verify the output.

Solution:

```
<?php
$subjects=["PHP","Java","DBMS"];
$marks=["PHP"=>85,"Java"=>78,"DBMS"=>90];
$students=[["name"=>"Amit","marks"=>82],["name"=>"Neha","marks"=>91]];
foreach($subjects as $s) echo "$s<br>";
foreach($marks as $s=>$m) echo "$s : $m<br>";
```

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```
foreach($students as $s) echo $s["name"]." : ".$s["marks"]."<br>";
?>
```

Expected Output:

Indexed: PHP, Java, DBMS | Associative: PHP 85, Java 78, DBMS 90 | Multidimensional: Amit 82, Neha 91

Result:

Thus, the practical was executed successfully and the required result was obtained.

Viva Questions:

What is an indexed array?

What is an associative array?

What is a multidimensional array?

Practical 6: Classes, objects and inheritance in PHP

Unit 3

Aim: To implement and demonstrate classes, objects and inheritance in php.

Procedure:

Start XAMPP and Apache.

Create practical6.php inside C:\xampp\htdocs\.

Enter the program, save it, and open http://localhost/practical6.php in the browser.

Verify the output.

Solution:

```
<?php
class Student{
public $name,$rollNo;
function __construct($n,$r){$this->name=$n;$this->rollNo=$r;}
function display(){echo "Name: $this->name<br>Roll No: $this->rollNo<br>";}
}
class Result extends Student{
public $marks;
function __construct($n,$r,$m){parent::__construct($n,$r);$this->marks=$m;}
function show(){ $p=array_sum($this->marks)/count($this->marks); $this->display(); echo "Percentage: $p%"; }
}
$s=new Result("Neha",101,[80,85,90]); $s->show();
?>
```

Expected Output:

Name: Neha | Roll No: 101 | Percentage: 85%

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Result:

Thus, the practical was executed successfully and the required result was obtained.

Viva Questions:

What is a class?

What is an object?

What is inheritance?

Practical 7: HTML form handling and validation using PHP

Unit 4

Aim: To implement and demonstrate html form handling and validation using PHP.

Procedure:

Start XAMPP and Apache.

Create practical7.php inside C:\xampp\htdocs\.

Enter the program, save it, and open http://localhost/practical7.php in the browser.

Verify the output.

Solution:

```
<!DOCTYPE html><html><body>
<form method="post">
Name: <input name="name" required><br>
Email: <input type="email" name="email" required><br>
Age: <input type="number" name="age" min="18" max="60" required><br>
<input type="submit" name="submit" value="Register">
</form>
<?php
if(isset($_POST['submit'])){
$name=trim($_POST['name']??''); $email=trim($_POST['email']??''); $age=(int)($_POST['age']??0);
if($name===''||!filter_var($email,FILTER_VALIDATE_EMAIL)||$age<18||$age>60) echo 'Please enter
valid details.';
else echo 'Registration Successful!<br>Name: '.htmlspecialchars($name).'<br>Email:
'.htmlspecialchars($email).'<br>Age: '.$age;
}
?></body></html>
```

Expected Output:

Entered: Neha | neha@example.com | 22 | Registration Successful!

Result:

Thus, the practical was executed successfully and the required result was obtained.

Viva Questions:

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What is $_POST?

Why use htmlspecialchars()?

What does filter_var() do?

Practical 8: PHP application to create dynamic images with text, buttons and scaling

Unit 4

Aim: To implement and demonstrate PHP application to create dynamic images with text, buttons and scaling.

Procedure:

Start XAMPP and Apache.

Create practical8.php inside C:\xampp\htdocs\.

Enter the program, save it, and open http://localhost/practical8.php in the browser.

Verify the output.

GD requirement: enable extension=gd in C:\xampp\php\php.ini if it is disabled, then restart Apache.

Solution:

```
<?php
header("Content-Type: image/png");
$image=imagecreatetruecolor(600,350);
$bg=imagecolorallocate($image,245,245,245);
$blue=imagecolorallocate($image,30,100,200);
$white=imagecolorallocate($image,255,255,255);
imagefill($image,0,0,$bg);
imagestring($image,5,180,70,"MCA PHP Practical",$blue);
imagefilledrectangle($image,200,150,400,210,$blue);
imagestring($image,5,260,172,"CLICK",$white);
$scaled=imagescale($image,300,175);
imagepng($scaled); imagedestroy($image); imagedestroy($scaled);
?>
```

Expected Output:

Browser displays a generated 300 x 175 PNG with text and a CLICK button, scaled from 600 x 350.

Result:

Thus the practical was executed successfully and the required result was obtained.

Viva Questions:

What is the GD library?

What does imagecreatetruecolor() do?

What does imagescale() do?

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